

Characteristics of data & information

Six properties decide whether a dataset is actually fit for the job : size, structure, relevance, accessibility, clarity and context. Get them right and raw data becomes useful information. Get them wrong and even "correct" data is useless.

What this key knowledge covers

This summary distils everything you need for **KK04 — Characteristics of data & information** in Data Analysis. Work through the explanations, then use the worked good-vs-weak answers to see exactly how marks are won and lost. Tick off the revision checklist at the end before a SAC or exam.

Study summary

First: data is not information

You cannot describe the characteristics of "data and information" until you can tell the two apart. The study design lists them together for a reason — **context is the bridge between them.**

Data

Raw, unprocessed facts and figures with no meaning on their own. A value, a string, a timestamp. 237411 is data. So is "Skip".

Information

Data that has been processed, organised and *given context* so it answers a question and supports a decision. "You listened for 4 hrs less in winter" is information.

Try it. Below is a raw value from Maya's export. Add context and watch data become information:

The six characteristics — see each one

Now the detail — definition, what it looks like in Maya's SoundCheck file, and the question each one answers:

Structure: the three states of data

Structure is the characteristic students lose the most marks on, because it has named categories examiners expect you to use:

The "relevance" trap: KK03 vs KK04

Relevance appears in *both* dot points, and so does the word "information". Students mix them up under exam pressure. They are different lists answering different questions:

Memory hook: **quality is about whether the data is right; characteristics are about whether the data is right for you.**

Sort the situations

Each card describes something that went wrong (or right) with Maya's data. Drag it to the characteristic it best demonstrates. On a touchscreen, tap a card then tap a zone.

How to write a characteristics answer: P · E · L

Extended-response marks are lost when students name a characteristic but never connect it to the scenario. Use **P·E·L** for every characteristic you discuss:

Common exam traps

- × Listing **quality factors** (accuracy, bias, reliability) when asked for **characteristics**. Wrong list.
- × Defining a characteristic but never linking it to the scenario — that's the missing mark almost every time.
- × Treating **size** as "always more is better". Too much data is slow and noisy; too little is unrepresentative. Describe the effect both ways.
- × Confusing **accessibility** (can the right people get and use it?) with **security** (keeping the wrong people out). They pull in opposite directions.
- × Saying data "is information". Until it is processed and given **context**, it is only data.
- × Mixing up **clarity** (is it understandable?) with **accuracy** (is it correct?). Data can be perfectly clear and still wrong.

Common traps & misconceptions

Exam trap

⚠ **If a question says "characteristics", do not answer with accuracy, bias, integrity or reliability.** Those are *quality factors* (KK03). The only word that legitimately belongs to both lists is **relevance** — and even then, frame it for the list the question asks about.

How to answer exam questions

Read the **command word** first — it sets how much to write. *State/Identify* wants a word or phrase; *Describe* wants features; *Explain* wants reasons and links; *Justify/Evaluate* wants a judgement backed by evidence. Always pull in the **scenario detail** rather than answering in the abstract. The examples below contrast a weak response with a strong one for the same question.

Q1. Define · 1 mark

Define what is meant by the “context” of data.

✗ A weak answer

Context is the background of the data. Spot what's missing before you reveal the strong answer.

✓ A strong answer

Context is the surrounding circumstances — such as when, where, from whom and measuring what — that give a piece of data meaning and turn it into usable information.

How the marks are awarded

- 1 mark: a definition that links context to the circumstances/meaning that make data usable.
- “Background” alone is too vague — it must convey that context provides *meaning*.

Q2. Distinguish · 2 marks

Using an example from Maya's SoundCheck export, distinguish between data and information.

✗ A weak answer

Data is small and information is big. Maya has lots of data. Spot what's missing before you reveal the strong answer.

✓ A strong answer

Data is raw, unprocessed facts — e.g. the value 237411 in Maya's ms_played column. Information is that data processed and given context — e.g. “Maya listened to ‘Glass’ for about 4 minutes on 3 July” — which now has meaning and can support a decision.

How the marks are awarded

- 1 mark: correct distinction (data = raw/unprocessed facts; information = processed + given context/meaning).
- 1 mark: a valid example of each drawn from the scenario.
- Size is *not* the difference — that's a common error.

Q3. Describe · 2 marks

Maya's export contains 52,000 rows. Referring to the characteristic of size, describe one benefit and one drawback this could create for her analysis.

✗ A weak answer

Having 52,000 rows is good because more data is always better. Spot what's missing before you reveal the strong answer.

✓ A strong answer

Benefit: a large dataset (52,000 rows) is likely to be representative, so Maya can reliably spot seasonal listening patterns. Drawback: that volume may be slow to open and filter in basic spreadsheet software and can contain more noise, making the analysis harder to manage.

How the marks are awarded

- 1 mark: a valid benefit of large size tied to the scenario (representativeness / reliable patterns).
- 1 mark: a valid drawback (performance / handling / noise).
- “More is always better” scores 0 — size has trade-offs in both directions.

Q4. Explain · 4 marks

Describe two characteristics of data and information (other than size) and, for each, explain how it could affect the quality of Maya's final visualisation.

✗ A weak answer

Accuracy — the data must be correct. Reliability — the data must be dependable. Both make the visualisation better. Spot what's missing before you reveal the strong answer.

✓ A strong answer

Structure: Maya's play events are structured into rows and columns, so she can sort and SUM listening time per genre directly — if part of her data were unstructured free text she'd have to reformat it first, delaying the build. Clarity: her ms_played field has no stated unit; if she doesn't relabel it as “Minutes listened”, axis labels on her chart will be ambiguous and the audience may misread the visualisation.

How the marks are awarded

- Up to 2 marks per characteristic: 1 for correctly naming + defining a KK04 characteristic, 1 for an effect on the visualisation tied to the scenario.
- Must be **characteristics** (structure, relevance, accessibility, clarity, context) — the weak answer used **quality factors** (accuracy, reliability) and earns no marks here.
- Two characteristics required; a single very detailed one caps at 2.